

Raging Bits ESP32 WS2812 8 CHAR DISPLAY v1.0

Top level specs

ESP32-WROOM-16Mbit central Module

8 char display with full RGB 5x7 matrix digit (ws2812 leds)

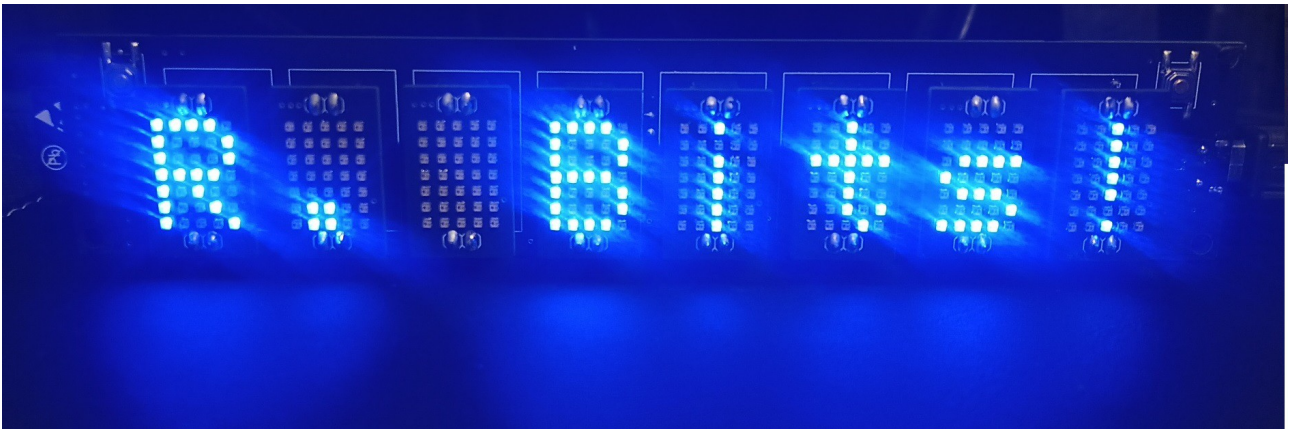
Battery backed up RTC. (CR1220 coin cell)

1 configurable Led.

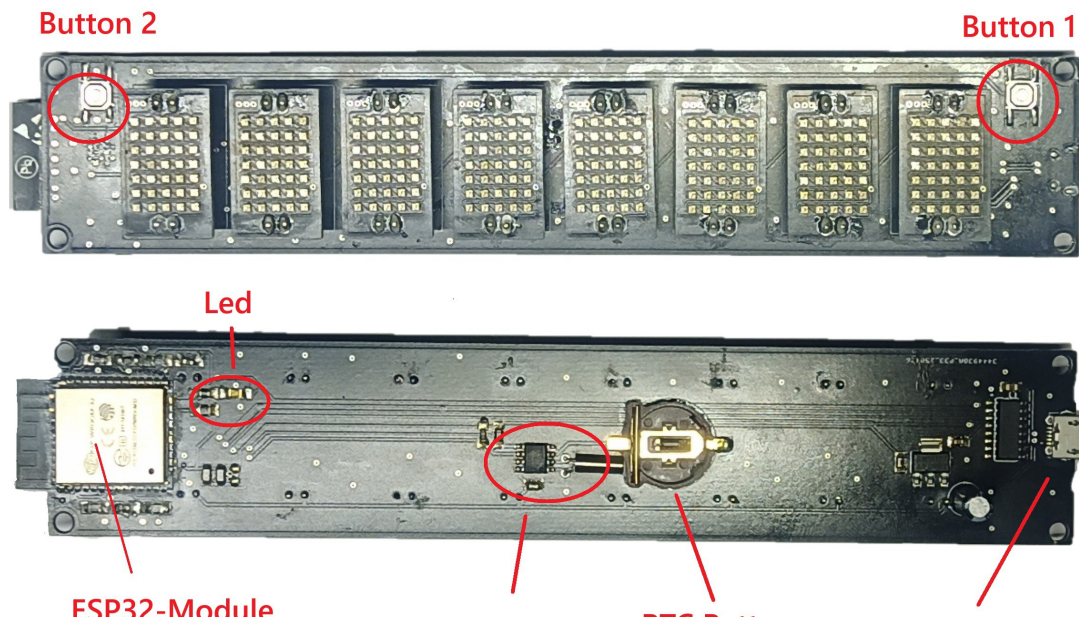
1 Serial Port (fixed 115200bps)

2 buttons for clock control interface

Standard micro USB 5V power serial port CH430.



Device interface



Compatible with WLED and many other softwares.

Hardware

Power

Power consumption at 5V is kept within USB standard, by limiting the brightness.

ESP32 Pinout

WS2812 control:	Pin 27 - GPIO16
Button2:	Pin 37 - GPIO23
Button1:	Pin 10 - GPIO25
Info LED:	Pin 24 - GPIO2
I2C SDA:	Pin 33 - GPIO21
I2C SCL:	Pin 36 - GPIO22

Programmer

The ESP is programmed directly from serial port.
The Port is fully wired oriented towards ESP tools programmers.

ESP32 programmer serial port

TX: Pin 35 - GPIO1

RX: Pin 34 - GPIO3
RTS: Pin 3 – ENABLE (Reset)
DTR: Pin 25 – GPIO0 (Boot)

Uart

For the Default application, the UART is fixed at 115200bps, 1 start bit, 1 stop bit, no flow control and no parity.

INFO LED

A simple LED can be set or reset by serial commands.

Buttons

The buttons are used to directly interface and make use of the display as a clock/calendar.

Button 1 click

A button 1 click will simply move the text to another colour.

There are 11 preset colours to choose from:

- Rainbow - keeps the text moving through all the colours.

- Red - solid

- Orange - solid

- Yellow - solid

- Green - solid

- Light Cyan - solid

- Cyan - solid

- Blue - solid

- Purple - solid

- Pink - solid

- White - solid

Button 2 click

A button 2 click will simply move through the clock date display options.

There are 4 different options available:

- Only time – No rotation.

- Only date – Rotating.

- Time and date – Rotating.

- Alternate Time and date – No rotation for time for 45s. Rotating date for 15s.

Button 1 hold+ Button 2 hold

When holding both buttons for over a second, the setup menu is invoked.

Once active, if no button presses are detected within 30s, the setup exits back to normal clock.

Setup Button 2

While in setup, button 2 controls values by incrementing by one unit with a click or continuously by holding.

Setup Button 1

While in setup, button 1 controls the settings menu items by incrementing by one unit with a click or continuously by holding.

Setup menu

The setup menu has 7 items and always goes top to bottom once before exiting:

- Set year – (2023 to 2099).

- Set month – (1 to 12).

- Set day – (1 to 31).

- Set hour – (0 to 23).

- Set minute – (0 to 59).

- Set brightness – (10 levels).

- Save – (yes or no).

Software

Serial port Commands

As mentioned before, the Default software allows for formatted commands at 115200bps. The system is designed to have the commands synchronized using a simple framing system. The frame starts with a Line Feed character, followed by the command identification, the command data if any, and ends with a Carriage Return. This means all command data will be in ASCII not binary.

[`\n`] [`Command ID`] [`Data (if any)`] [`\r`]

Show Text – '1' (0x31)

Every new text entry resets the display rotation point. Due to the nature of the HPDL1414, only a limited set of characters are supported, Numbers, only Upper Case letters and some signs. (There is a table with the full supported contents in the Notes.)

[`\n`] [`1`] [`Text to show`] [`\r`]

```
Serial > n1HELLO WORLD    \r
Serial < \nOK\r
```

Display now rotates the message "HELLO WORLD ".

Show Time – '2' (0x32)

[`\n`] [`2`] [`\r`]

```
Serial > n2\r
Serial < \nOK\r
```

Display will now show the time.

Show Time Date – '3' (0x33)

[`\n`] [`3`] [`\r`]

```
Serial > n3\r
Serial < \nOK\r
```

Display will now show the time and date rotating.

Get Time-Date – '4' (0x34)

[`\n`] [`4`] [`\r`]

```
Serial > n4\r
```

Serial < \n00:10:13 01 FEB 2023\r

Display will return the time and date.

Set Time-Date – '5' (0x35)

The time and date format MUST BE EXACTLY as follow.

hh:mm:ss dd/MM/yyyy

where:

hh – 2 digit hours
mm – 2 digit minutes
ss – 2 digit seconds
dd – 2 digit day
MM – 2 digit month
yyy – 4 digit year

[\n] [5] [hh:mm:ss dd/MM/yyyy] [\r]

Serial > \n500:10:13 01/02/23\r

Serial < \nOK\r

Display will return the time and date.

Set Work LED – '6' (0x36)

[\n] [8] [Status] [\r]

Serial > \n60\r

Serial < \nOK\r

Led is now OFF.

Serial > \n81\r

Serial < \nOK\r

Led is now ON.

Set Display text colour – '7' (0x37)

Sets the text colour with either a solid colour or a progressive rainbow.

The rainbow effect is chosen by having the effect data as 0.

The solid colour is chose nby having the effect data as 1 and will admit a colour setup of a combination of 10 gradients per component of the RGB.

[\n] [7] [(0/1)Effect] [R(0-9)] [G(0-9)] [B(0-9)] [\r]

Serial > \n70000\r

Serial < \nOK\r

Sets Rainbow colours.

Serial > \n71999\r
Serial < \nOK\r

Serial > \n71555\r
Serial < \nOK\r

Led Solid white

Serial > \n71900\r
Serial < \nOK\r

Led Solid red

Set Display text brightness – '8' (0x38)

The text brightness command will set the intensity of the display brightness. A simple factor over the colour. The brightness can be set between 0 and 100.

[\n] [7] [(0 - 1)] [(0 - 9)] [(0 - 9)] [\r]

Serial > \n8000\r
Serial < \nOK\r

Brightness set to zero

Serial > \n8058\r
Serial < \nOK\r

Brightness set to 58

Notes

The Software information of this datasheet is only valid for the Default code that is shipped in the device. The Default original code will always be available for reflash by the user.

Disclaimer:

No guarantees are given for fully compatibility with other codes.

The device hardware was designed to be compatible with the WLED software (and others) although this is only true at the moment of the device first version being released.

To program the device with WLED, all that is needed is to go to WLED page project and use the programming webpage. The whole process should be fully automatic.

The leds are ws2812.

The 2D matrix organisation configuration for WLED should be:

8 Tiles total, 1 vertical x 8 horizontal, organized from left to right.

Each tile starts from top left to bottom right in zigzag.



Others:

